

ITC-101

Examination –Dec- 2023

B.Tech. II Sem : CE, AE, EE, EI, EC, CS, IT, AI, BC, IoT

Python Programming

Max. Marks :60

Min. Marks :19

Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a & b – Objective type/Fill in the blanks/One sentence answer.

Part c – Max 100 words and Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

		Marks	BL	CO
(a)	1. Who developed Python Programming Language? (i) Wick van Rossum (ii) RasmusLerdorf (iii) Guido van Rossum (iv) NieneStom 2. Which type of Programming does Python support? (i) object-oriented programming (ii) structured programming (iii) functional programming (iv) all of the mentioned	02	2	1
(b)	1. Is Python case sensitive when dealing with identifiers? (i) no (ii) yes (iii) machine dependent (iv) none of the mentioned 2. Which of the following is the correct extension of the Python file? (i) .python (ii) .pl (iii) .py (iv) .p	02	2	1
(c)	What are the features of Python programming?	03	2	1
OR				
	How python programming is different from the other programming languages?	03	2	1
(d)	Write about python data types in details.	05	1	1
OR				
	What is meant by indentation and comments in python programming languages? Explain it with the help of an example.	05	2	1
(a)	1. What will be the output of the following Python code? i=1 while True: if i%3==0: break print(i) i +=1 (i) 1 2 3 (ii) error (iii) 1 2 (iv) none of the mentioned 2. Keyword “and” is a logical operator(True/False).	02	3	2

(b) 1. What will be the output of the following Python code?

```
i = 0
while i < 3:
    print(i)
    i += 1
else:
```

* print(0)

(i) 0 1 2 3 0 (ii) 0 1 2 0 (iii) 0 1 2 (iv) error

2. Keyword "class" is used to define a class (True/False).

(c) Write about logical operators of python programming.

OR

Write about the Break statement of python programming.

(d) Write about the if, else if and nested if statement.

OR

Write a program to show the use of while loop and for loop.

Q.3 (a) 1. Python does not have a character data type, a single character is simply a string with a length of 1.(True/False)

2. The output of executing string.ascii_letters can also be achieved by:

(i) string.ascii_lowercase_string.digits
(ii) string.ascii_lowercase+string.ascii_uppercase
(iii) string.letters
(iv) string.lowercase_string.uppercase

(b) 1. Strings in Python can be created using single quotes or double quotes or even triple quotes. (True/False)

2. What arithmetic operators cannot be used with strings?
(i) + (ii) * (iii) - (iv) All of the mentioned

(c) How to create a string in python?

OR

What do you understand by string slicing?

(d) Write a program to show updating and deleting a string.

OR

What is Escape Sequencing in Python? Explain it with the help of an example.

Q.4 (a) 1. Which of the following commands will create a list?
(i) list1 = list() (ii) list1 = [] (iii) list1 = list([1, 2, 3]) (iv) all of the mentioned

2. What is the output when we execute list("hello")?
(i) ['h', 'e', 'l', 'l', 'o'] (ii) ['hello'] (iii) ['llo'] (iv) ['olleh']

b)	1. Which of the following is a Python tuple? (i) [1, 2, 3] (ii) (1, 2, 3) (iii) {1, 2, 3} (iv) {}	02	2	4
	2. Tuple is a collection of Python objects much like a list. (True/False)			
c)	What is the difference between tuples and lists in Python?	03	1	4
	OR			
	Write a few methods that are used in Python Lists.	03	1	4
d)	Explain how to create a dictionary in python?	05	3	4
	OR			
	How lists are updated in Python?	05	3	4
a)	1. _____ represents an entity in the real world with its identity and behaviour. (i) A method (ii) An object (iii) A class (iv) An operator	02	1	5
	2. _____ is used to create an object. (i) class (ii) constructor (iii) User-defined functions (iv) In-built functions			
b)	1. A class is a collection of objects. (True/False)	02	1	5
	2. What is getattr() used for? (i) To access the attribute of the object (ii) To delete an attribute (iii) To check if an attribute exists or not (iv) To set an attribute			
c)	What are classes and objects?	03	1	5
	OR			
	What is meant by methods and attributes?	03	2	5
d)	What the main features are of object oriented programming? Write in detail.	05	2	5
	OR			
	Explain the concept of overloading and overriding with the help of example.	05	3	5

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating
